

Lab 04

spice

Contents

Contents	2
Part 1 (prelab).....	5
1 Using ADT Device Xplore, plot the following design charts vs g_m/ID for both PMOS and NMOS. Set $V_{DS} = V_{DD}/3$ and $L = 0.18\mu, 0.5\mu:0.5\mu:2\mu$	5
1.1 Nmos	5
1.1.1 g_m/g_{ds}	5
1.1.2 ID/W	6
1.1.3 g_m/C_{gg}	6
1.1.4 V_{GS}	7
1.2 PMOS	7
1.2.1 g_m/g_{ds}	7
1.2.2 ID/W	8
1.2.3 g_m/C_{gg}	8
1.2.4 V_{GS}	9
2 Use ADT Device Xplore to find ID and V_{GS} of an NMOS that has $g_m/ID = 10$, $g_m/g_{ds} = 50$, $V_{DS} = V_{DD}/3$, $V_{SB} = 0$, and $W = 5\mu$	9
3 Complete the script “lookup_test.m” to repeat the previous problem. Compare the results in a table.....	11
4 Use ADT Device Xplore to design an NMOS-input 5T-OTA	12

4.1	gm/ID = 15 for the input pair and gm/ID = 10 for the load and the tail bias (why is this reasonable?).	12
4.2	ro of the load is five times ro of the input pair (why is this reasonable?).	12
4.3	L = 1um for the tail bias (why is this reasonable?).	12
4.4	DESIGN	13
4.5	using charts:	13
4.5.1	input pair:	13
4.5.2	Active Load:	15
4.5.3	Tail bias:	16
5	Complete the function “designOTA.m”, Report your MATLAB function.	17
6	Complete the script “designOTA_test.m”, Report your script. Report the output of the command window, Compare the results to the results obtained by ADT Device Xplore.	19
7	Complete the netlist “ota_tb.cir”, Simulate the differential gain vs frequency (annotate DC gain and GBW), Report your netlist, the simulation results, and the comparison table. Comment on the results.	21
PART2		25
1	Use ADT Device Xplore to redesign the previous OTA while taking into account the OTA self-loading , Compare the results to the results obtained in Part 1	25
1.1	COMPARISION WITH PART1:	30

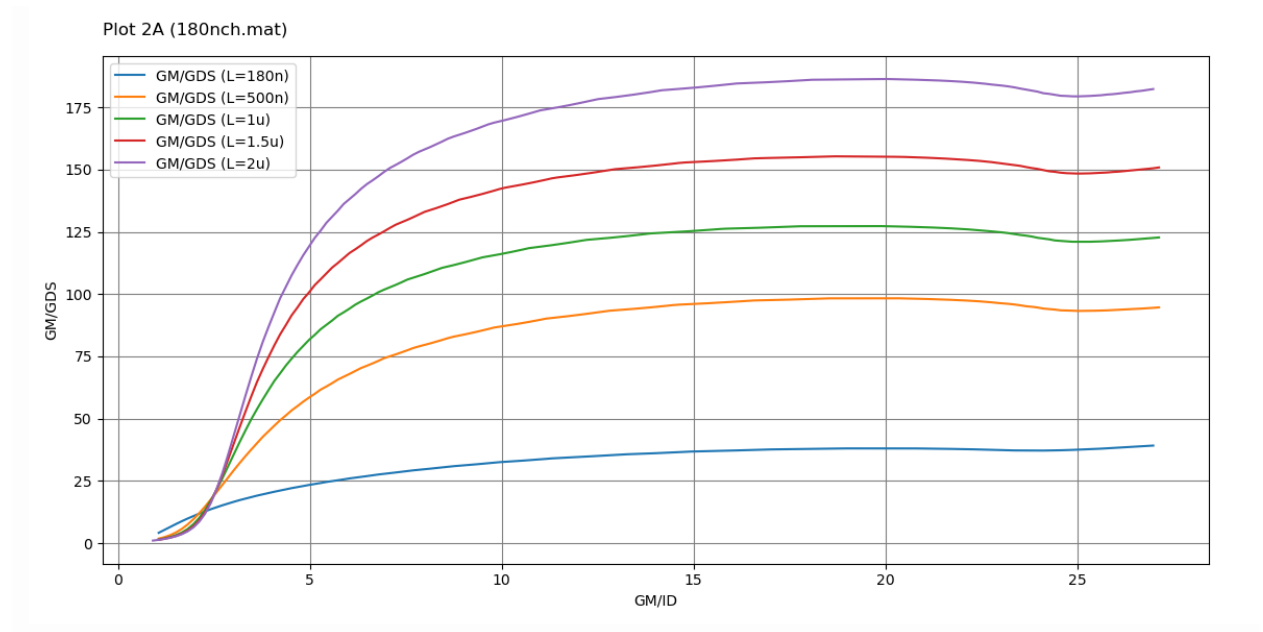
2	MATLAB SCRIPT,. Compare the results to the results obtained by ADT Device Xplore.....	31
2.1	Report the output of the command window	31
2.2	Compare the results to the results obtained in Part 1.	31
2.3	Compare the results to the results obtained by ADT Device Xplore.....	32
3	Write a netlist for the synthesized OTA. Simulate the differential gain vs frequency (annotate DC gain and GBW) . Comment on the results.	33
3.1	Report your netlists.....	33
3.2	the simulation results.....	34
3.3	comparison table	34

Part 1 (prelab)

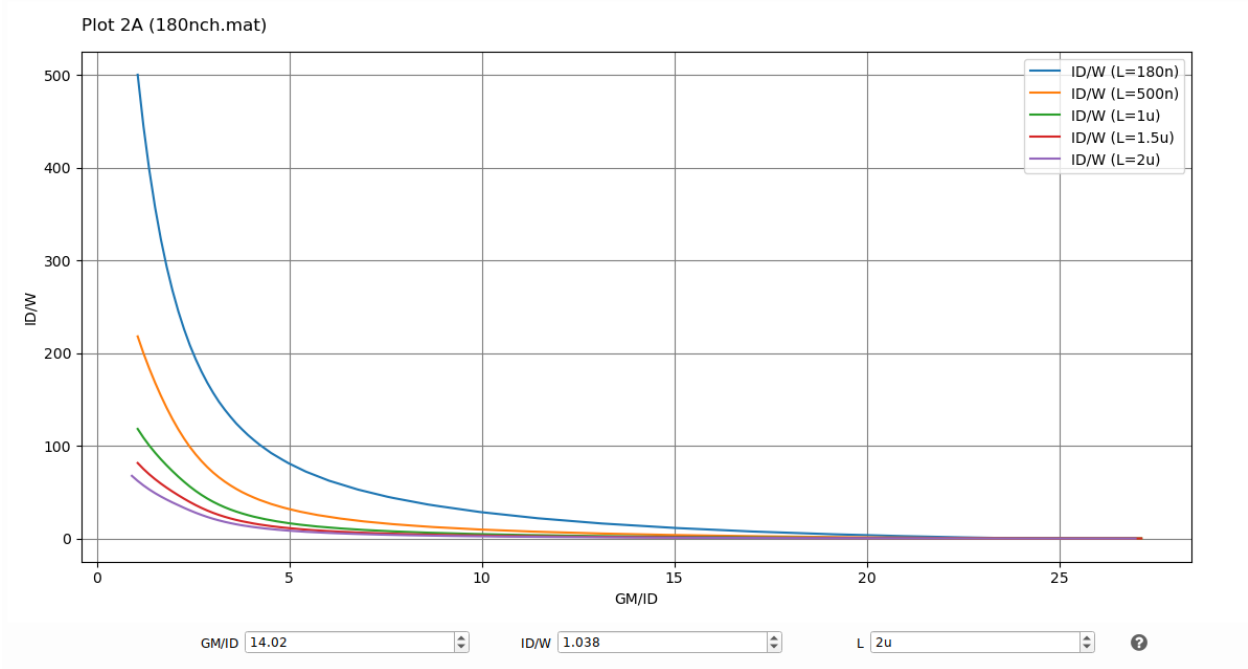
- 1 USING ADT DEVICE XPLORE, PLOT THE FOLLOWING DESIGN CHARTS VS GM/ID FOR BOTH PMOS AND NMOS. SET $V_{DS} = V_{DD}/3$ AND $L = 0.18\mu, 0.5\mu:0.5\mu:2\mu$

1.1 NMOS

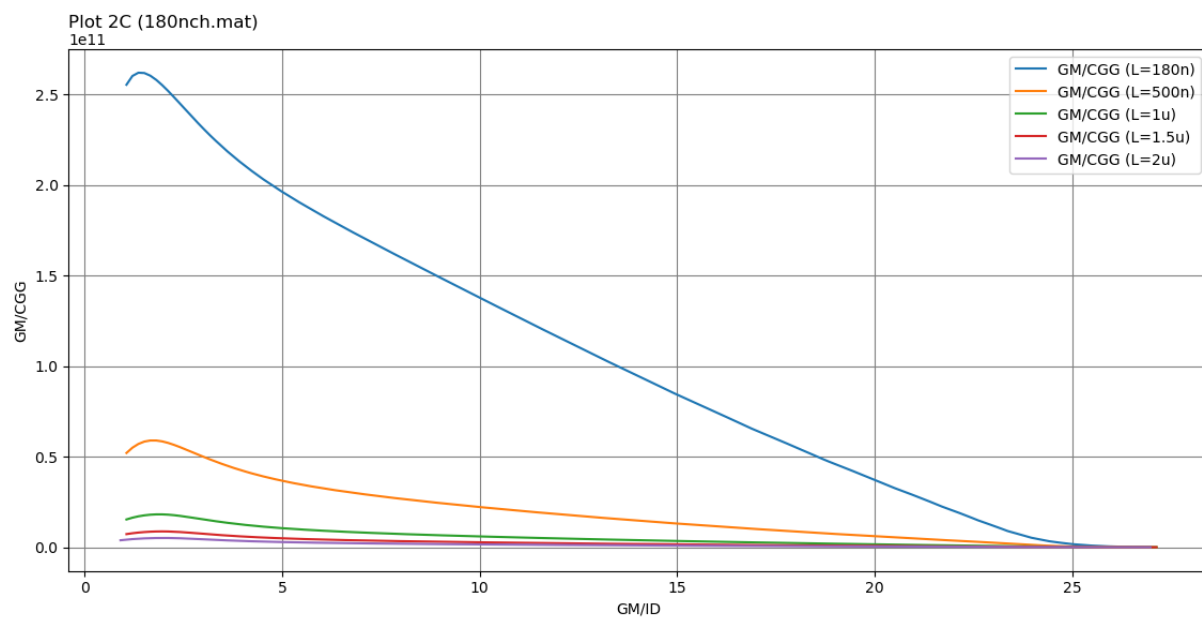
1.1.1 gm/g_{ds}



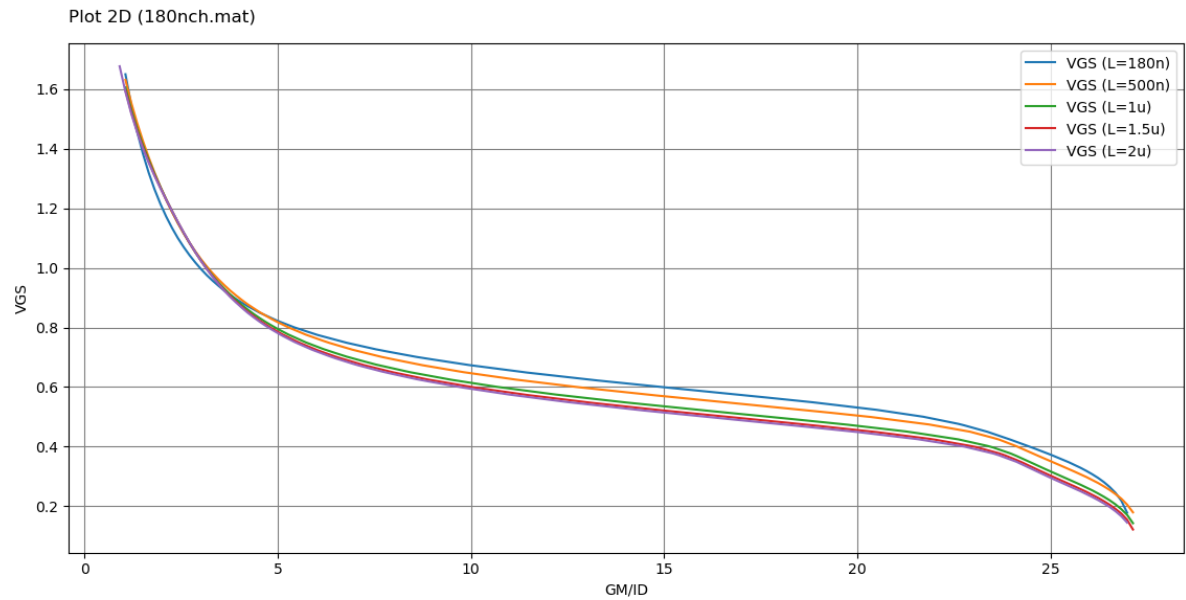
1.1.2 ID/W



1.1.3 gm/Cgg

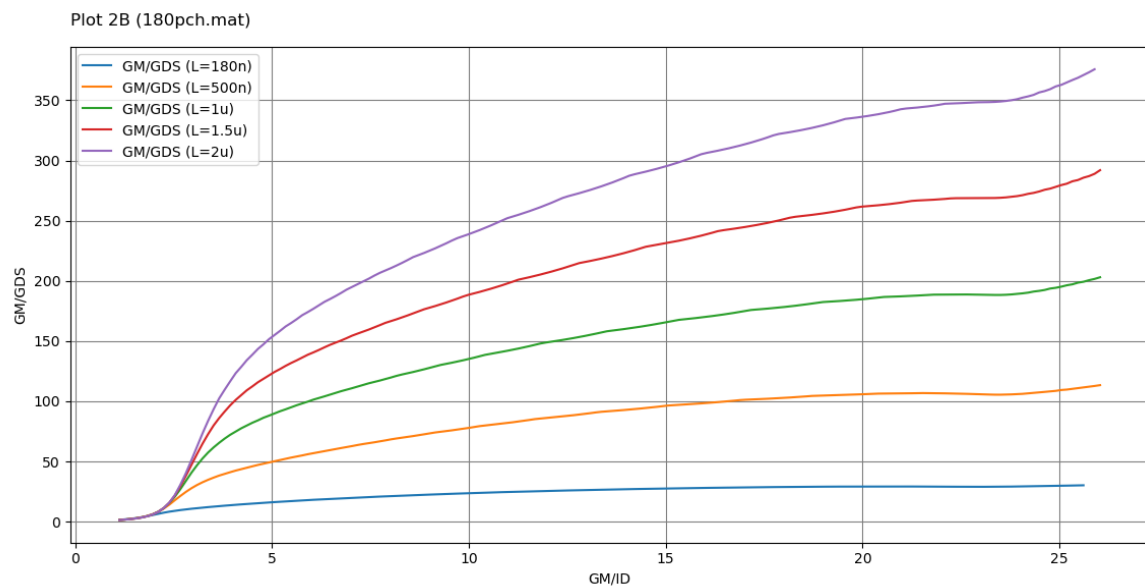


1.1.4 VGS

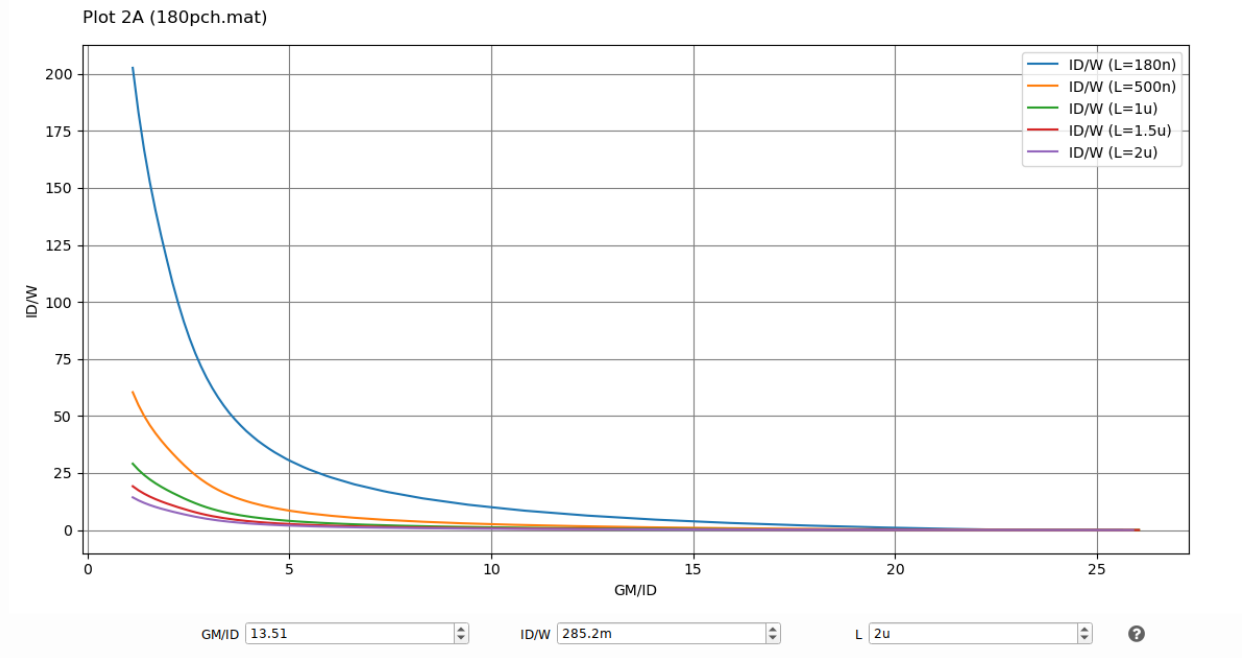


1.2 PMOS

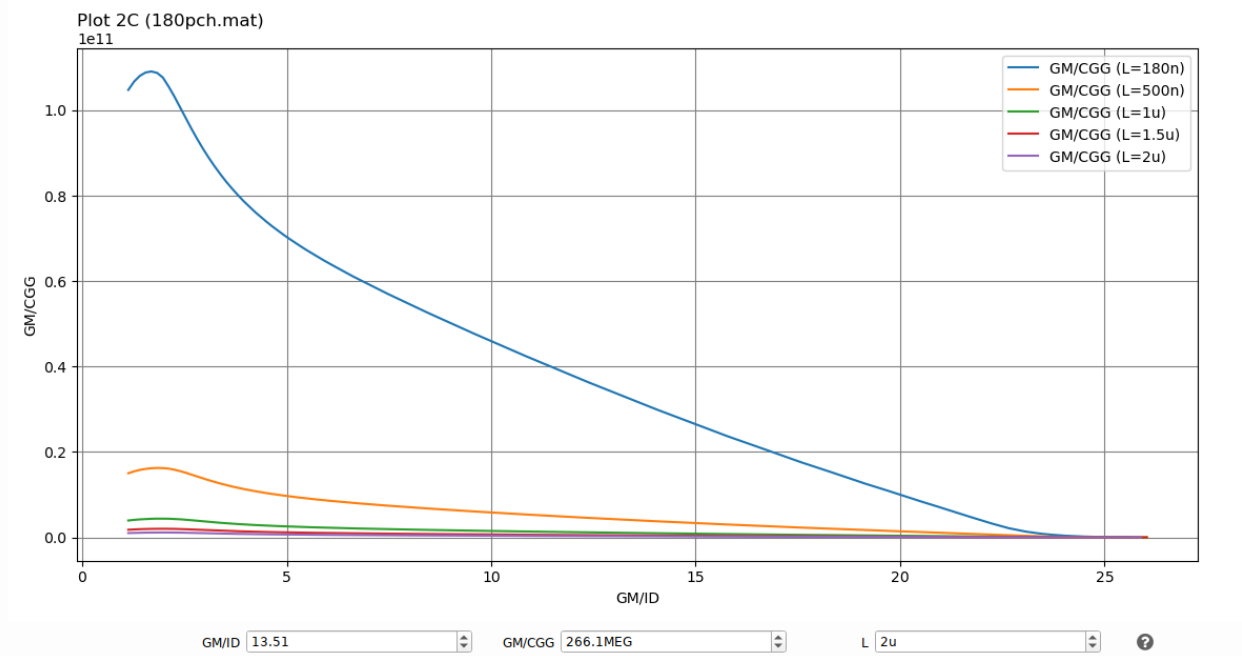
1.2.1 gm/gds



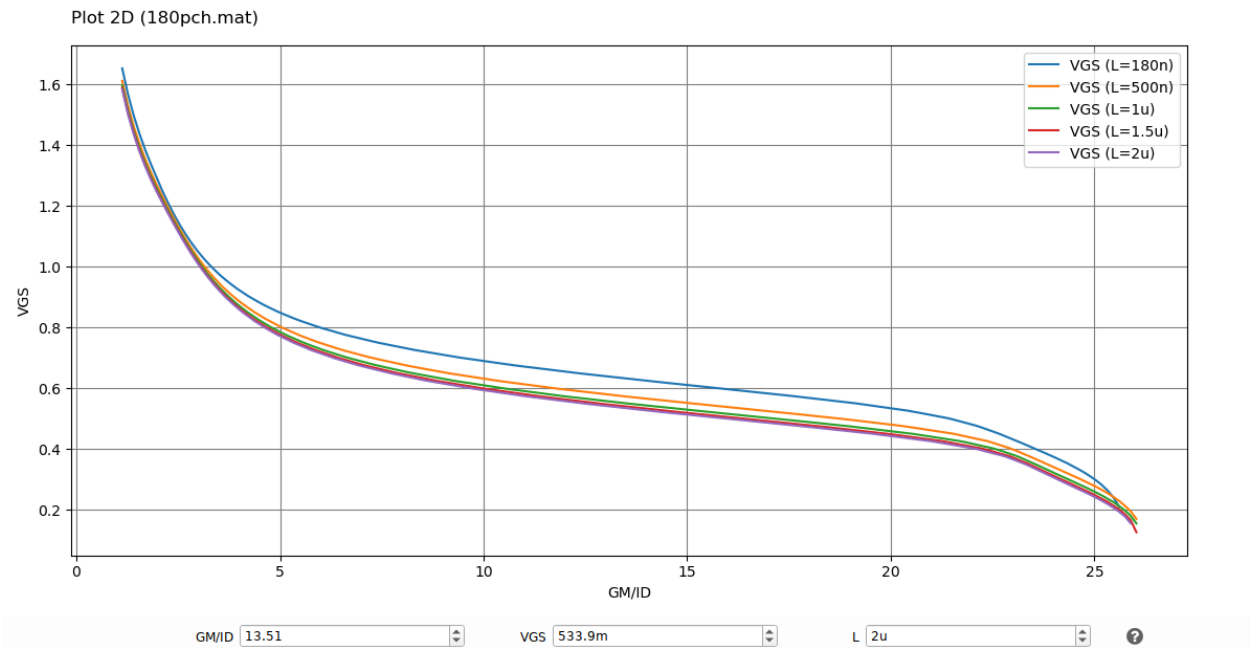
1.2.2 ID/W



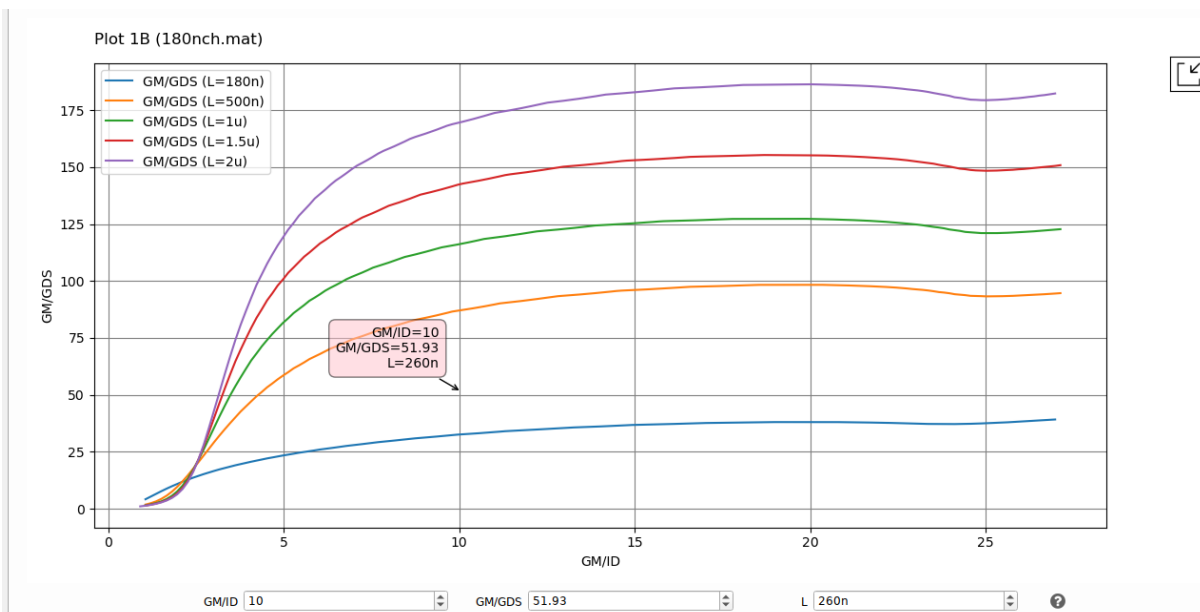
1.2.3 gm/Cgg



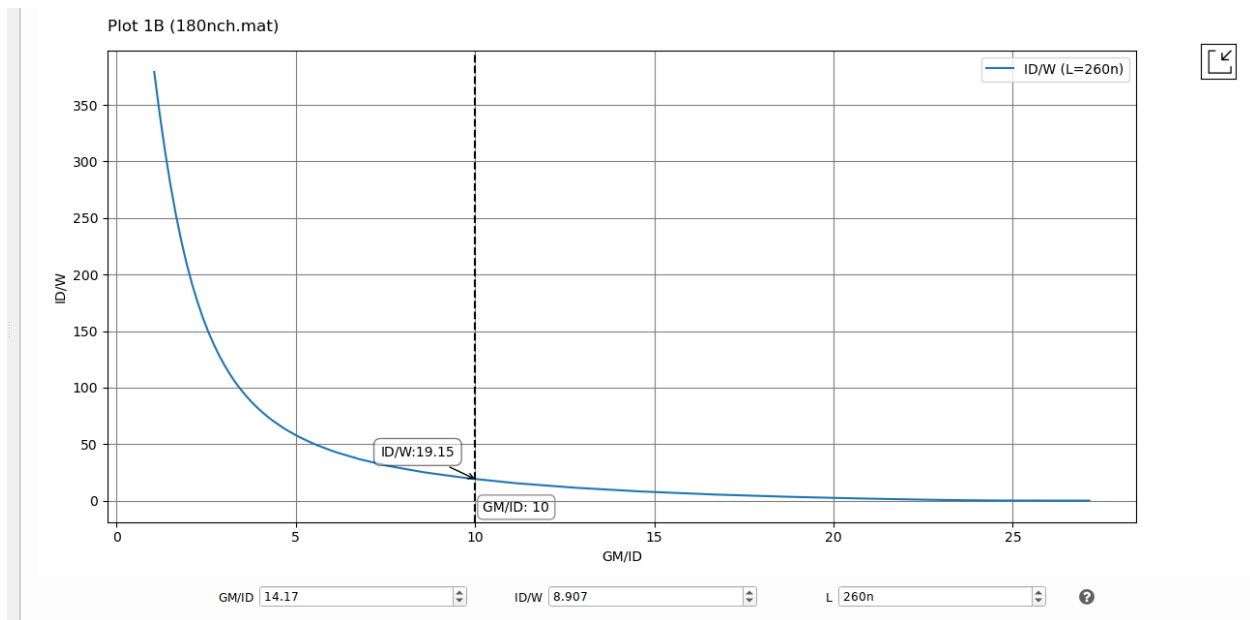
1.2.4 VGS



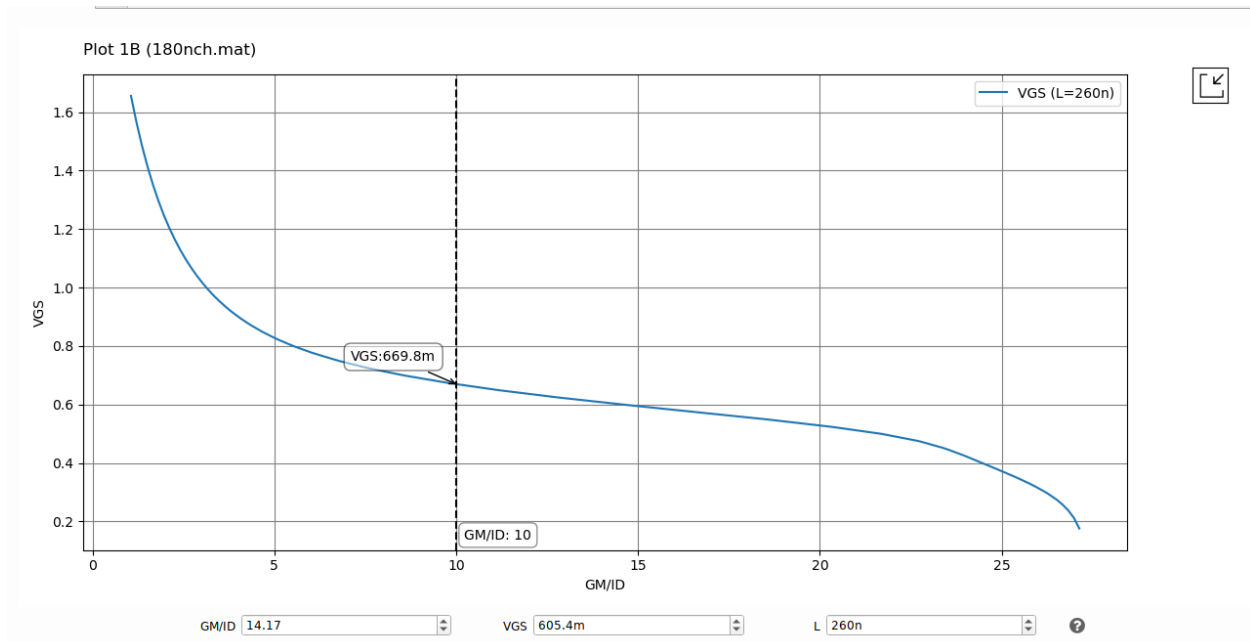
2 USE ADT DEVICE XPLORE TO FIND ID AND V_{GS} OF AN NMOS THAT HAS $GM/ID = 10$, $GM/GDS = 50$, $V_{DS} = V_{DD}/3$, $V_{SB} = 0$, AND $W = 5U$.



using magic cursor to get $GM/GDS > 50$ we chose $L = 260n$



At $gm/id=10$, $id/w=19.15$ and $W=5u$, so $ID=95.75u$



VGS=669.8m

3 COMPLETE THE SCRIPT “LOOKUP_TEST.M” TO REPEAT THE PREVIOUS PROBLEM. COMPARE THE RESULTS IN A TABLE.

```
% Lookup Test
% Please copy this script to gm/ID starter directory & run
it from there
% Clear workspace
clear all;

%load LUTs
load 180nch.mat;
load 180pch.mat;

% Compute VGS & ID
% Givens
VDD = 1.8;
M1.gm_ID = 10;
M1.gm_gds = 50;
M1.VDS = VDD / 3;
M1.VSB = 0;
M1.W = 5;
M1.gm_gds = 50;
% Find L that give gm/gds > given value
L_vector = nch.L; % Get the L vector from LUT structure
%Get the gm/gds values vector corresponding to the L_vector
gm_gds_vector = look_up(nch, 'GM_GDS', 'GM_ID', M1.gm_ID,
'VDS', M1.VDS, 'L', L_vector);
% Get the minimum L that gives gm/gds > the given value
M1.L=min(L_vector(gm_gds_vector>=M1.gm_gds)) ;
% Get the current by computing the ID/W and then multiply
it by W
M1.ID_W = look_up(nch, 'ID_W', 'GM_ID', M1.gm_ID, 'VDS',
M1.VDS, 'L', M1.L)
M1.ID=M1.ID_W*M1.W ;
% Get the VGS value
M1.VGS=look_upVGS(nch, 'GM_ID', M1.gm_ID, 'VDS', M1.VDS,
'L', M1.L)
% Print the solution
fprintf('VGS = %.2f\n',M1.VGS);
fprintf('ID = %.2d\n',M1.ID);
```

VGS = 0.67
ID = 9.52e-05

	MATLAB	ADT
VGS (mv)	670	670
ID (uA)	95.2u	95.75u

4 USE ADT DEVICE XPLORE TO DESIGN AN NMOS-INPUT 5T-OTA

4.1 $gm/ID = 15$ FOR THE INPUT PAIR AND $gm/ID = 10$ FOR THE LOAD AND THE TAIL BIAS (WHY IS THIS REASONABLE?).

As for the input pair we need bigger gm for more gain and more GBW as, so we bias it in MI, for the tail we biased it into SI so we would have large output resistance so smaller A_{vCM} , for the load we care more about the output resistance, so we bias it more towards the SI to get better λ for more gain. AS:

$$A_{diff} = -gm_{in} * (r_{in} // r_{load}) , GBW = \frac{gm_{in}}{2\pi CL}$$

4.2 RO OF THE LOAD IS FIVE TIMES RO OF THE INPUT PAIR (WHY IS THIS REASONABLE?).

as the parallel combination would result in better Rout so better gain, though it doesn't make sense to make r_{load} much bigger than r_{in} as the input pair not biased in SI so it has small λ_{in} so small r_o and the result of the parallel is smaller than r_o .

4.3 $L = 1\mu m$ FOR THE TAIL BIAS (WHY IS THIS REASONABLE?).

$$A_{vCM} = -\frac{1}{2gm_{load}R_{ss}}$$

We need large L for the tail so we would have larger R_{SS} so smaller A_{vCM} , we would have better CMRR

4.4 DESIGN

$$GBW = 100\text{Mhz} = \frac{gm_{in}}{2\pi CL}, gm_{in} = 315 \text{ uS}$$

$$\left(\frac{gm}{id}\right)_{in} = 15, ID = 21 \text{ uA}$$

$$A_V = \frac{5}{6} r_{in} gm_{in} = 50.2, r_{in} = 192 \text{ k}\Omega, r_{load} = 960 \text{ k}\Omega$$

$$\left(\frac{gm}{id}\right)_{load} = 10 = \frac{gm_{load}}{21\text{u}}, gm_{load} = 210 \text{ uS}$$

For the tail we assumed $L=1 \text{ um}$, $\frac{gm}{id} = 10$, and we know $I_{tail} = 42 \text{ uA}$, so $gm_{tail} = 420 \text{ uS}$

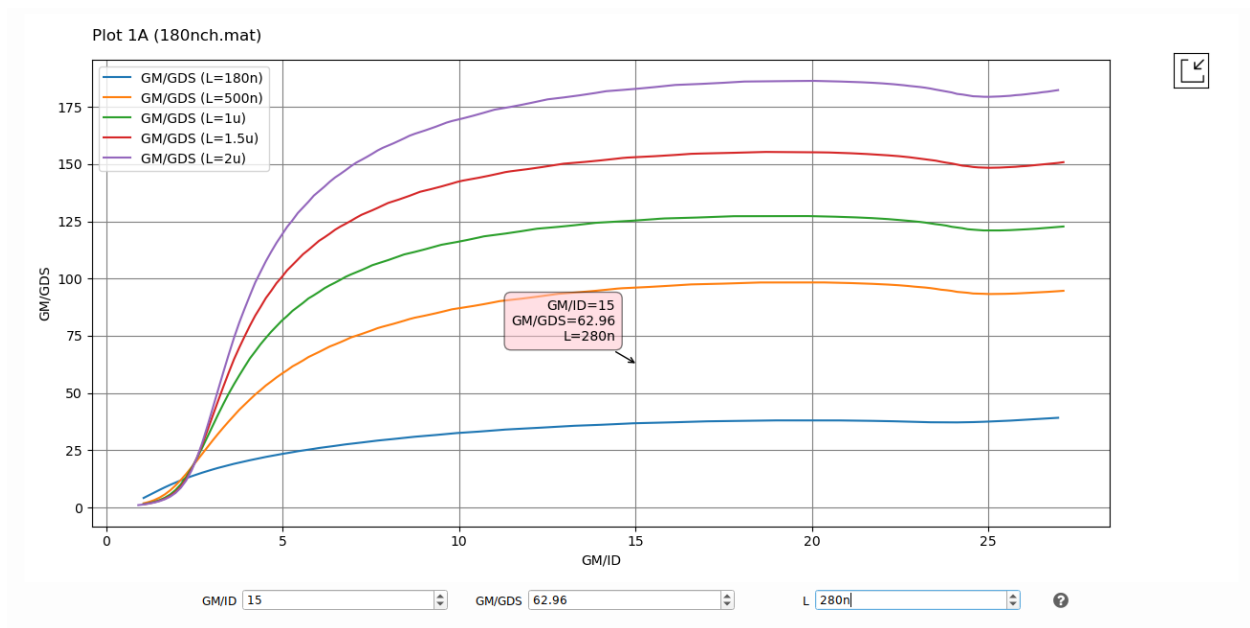
4.5 USING CHARTS:

4.5.1 input pair:

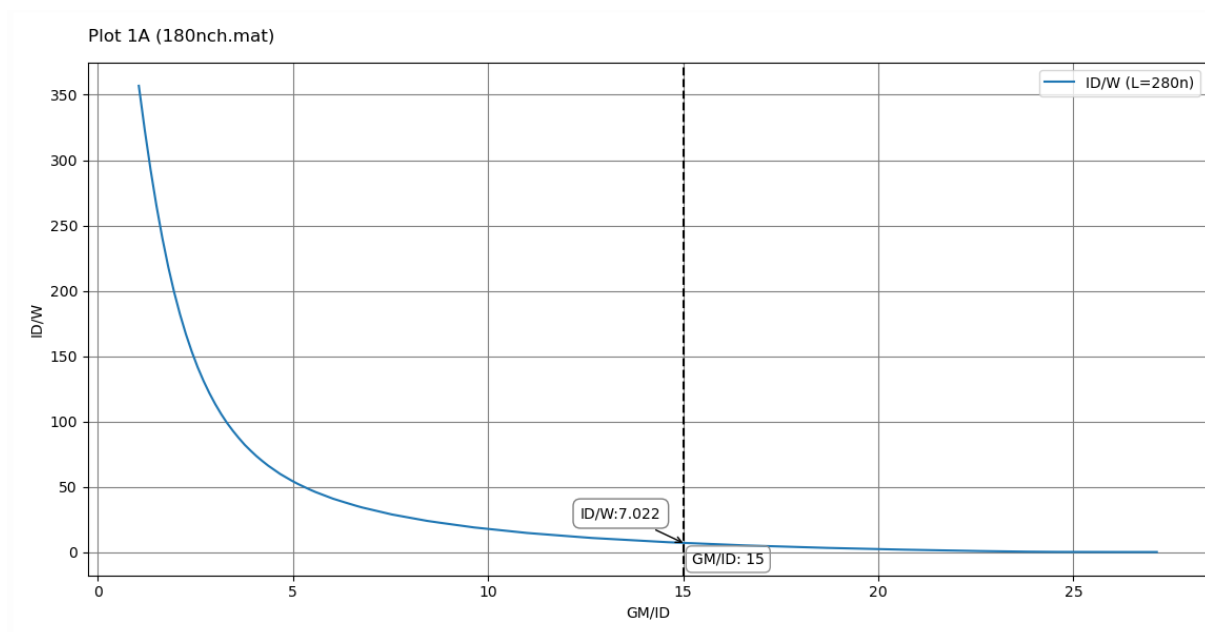
$$gm_{in} = 315 \text{ uS}, r_{in} = 192 \text{ k}\Omega$$

$$\left(\frac{gm}{gds}\right)_{in_{min}} = 60.48$$

$$\left(\frac{gm}{ID}\right)_{in} = 15$$



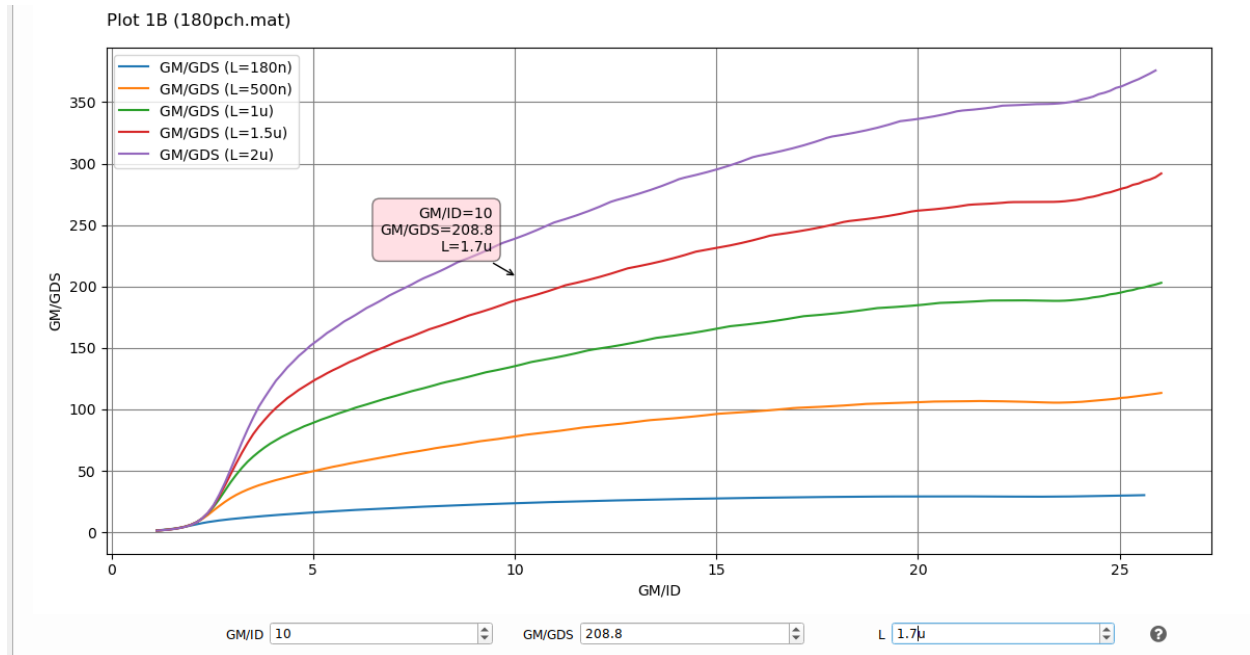
We choose **L=280n**, so $\frac{gm}{gds}=62.96$,



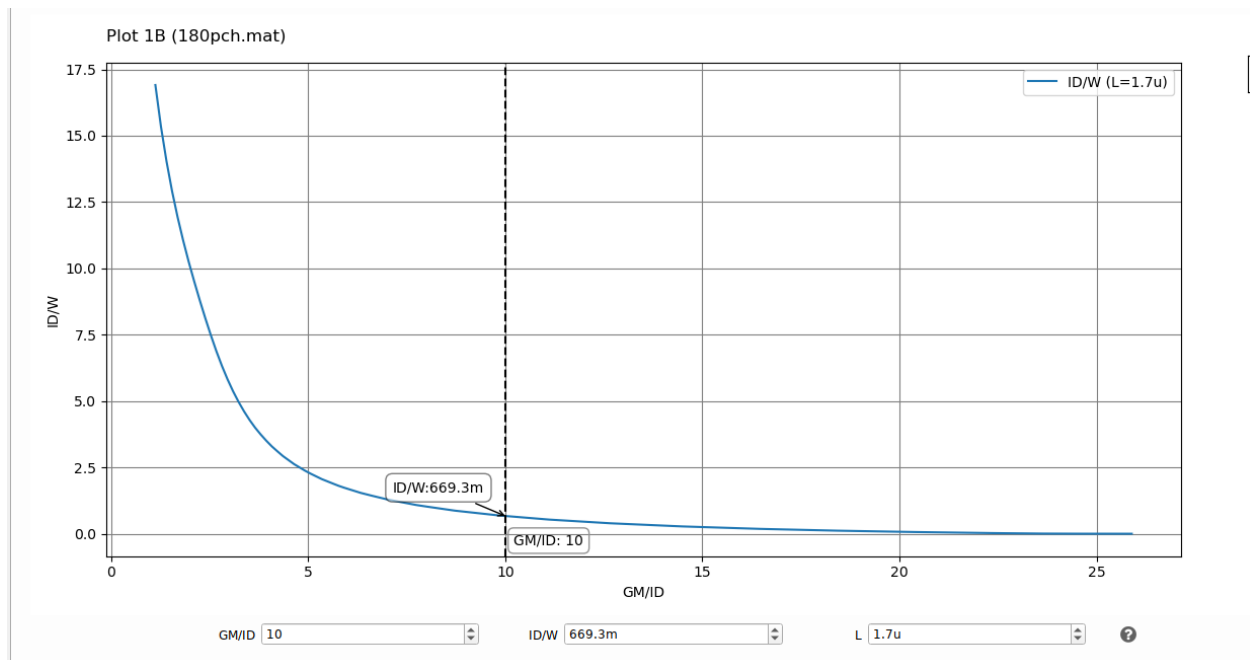
$$\frac{ID}{W} = 7.03, \text{ so } W = 3.01 \mu m$$

4.5.2 Active Load:

$$\left(\frac{gm}{gds}\right)_{load_{min}} = 201.6, \left(\frac{gm}{id}\right) = 10$$



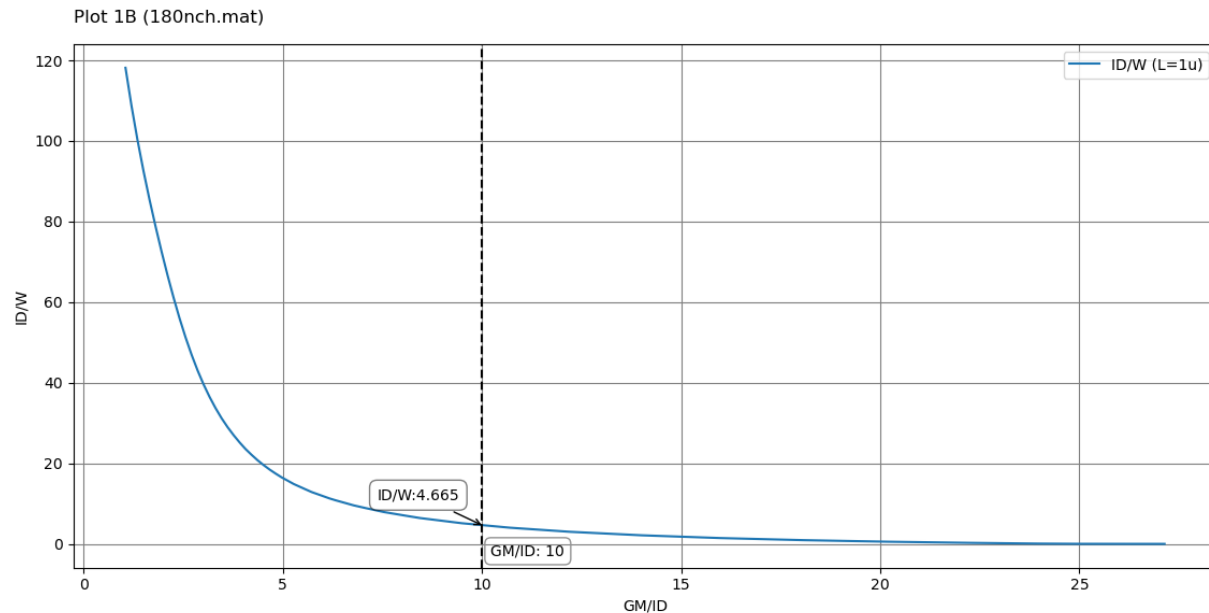
L=1.7u



$$\frac{ID}{W} = 669.3\mu A, \text{ so } W = 31.38 \mu m$$

The load trans has large size, but we didn't consider the parasitics until now in this part.

4.5.3 Tail bias:



$$L=1\mu m,$$

$$ID=42\mu A,$$

$$\frac{gm}{ID} = 10$$

$$\frac{ID}{W} = 4.665, \text{ so } W = 9 \mu m$$

5 COMPLETE THE FUNCTION “DESIGNOTA.M”, REPORT YOUR MATLAB FUNCTION.

```
%OTA Synthesis Function
function OTA = designOTA(specs)
%Additional Specs
VDD = 1.8;

%load LUTs
load 180nch.mat;
load 180pch.mat;

%Input Pair
OTA.M1.gm = specs.GBW * specs.CL * 2 * pi;
% assume ro(load) = 5 * ro(input) --> ro(input) = 6/5
R_total
DC_Gain_mag = 10^(specs.AVDC / 20); % Convert from dB to
mag
Rout = DC_Gain_mag / OTA.M1.gm; % Compute the equivalent
output resistance of OTA (Rout)

OTA.M1.ro = (6/5)*Rout % Complete the line to compute the ro
of M1

OTA.M1.gds = 1 / OTA.M1.ro;
OTA.M1.VDS = VDD/3;
OTA.M1.gm_gds = OTA.M1.gm / OTA.M1.gds;
OTA.M1.gm_ID = 15; % assumption

OTA.M1.ID = OTA.M1.gm / OTA.M1.gm_ID % Complete the line to
get the current of M1

% Search for the minimum L that gives gm / gds > specified
value
L_vector = nch.L;
gm_gds_vector = look_up(nch, 'GM_GDS', 'GM_ID',
OTA.M1.gm_ID, 'VDS', OTA.M1.VDS, 'L', L_vector);
OTA.M1.L = min(L_vector(gm_gds_vector >= OTA.M1.gm_gds)) ;
% Compute ID/W to get the W value
OTA.M1.ID_W = look_up(nch, 'ID_W', 'GM_ID', OTA.M1.gm_ID,
'VDS', OTA.M1.VDS, 'L', OTA.M1.L);
OTA.M1.W = OTA.M1.ID / OTA.M1.ID_W;
```

```

%CM_Load
OTA.M3.ID = OTA.M1.ID;

OTA.M3.ro = 5*OTA.M1.ro

OTA.M3.gds = 1 / OTA.M3.ro;
OTA.M3.VDS = VDD/3;
OTA.M3.gm_ID = 10;
OTA.M3.gm = OTA.M3.gm_ID * OTA.M3.ID;
OTA.M3.gm_gds = OTA.M3.gm / OTA.M3.gds;
gm_gds_vector = look_up(pch, 'GM_GDS', 'GM_ID',
OTA.M3.gm_ID, 'VDS', OTA.M3.VDS, 'L', L_vector);
OTA.M3.L = min(L_vector(gm_gds_vector > OTA.M3.gm_gds));
OTA.M3.ID_W = look_up(pch, 'ID_W', 'GM_ID', OTA.M3.gm_ID,
'VDS', OTA.M3.VDS, 'L', OTA.M3.L);
OTA.M3.W = OTA.M3.ID / OTA.M3.ID_W;

% Tail bias
OTA.M5.L = 1; %assumption
OTA.M5.ID = 2 * OTA.M1.ID;
OTA.M5.VDS = VDD/3;
OTA.M5.gm_ID = 10; %assumption
% Get ID/W to compute W
OTA.M5.ID_W = look_up(nch, 'ID_W', 'GM_ID', OTA.M5.gm_ID,
'VDS', OTA.M5.VDS, 'L', OTA.M5.L);
OTA.M5.W = OTA.M5.ID / OTA.M5.ID_W;

% get CMIN bias value
OTA.M1.VGS =look_upVGS(nch, 'GM_ID', OTA.M1.gm_ID, 'VDS',
OTA.M1.VDS, 'L', OTA.M1.L)
OTA.M1.VG = OTA.M1.VGS + OTA.M1.VDS

```

6 COMPLETE THE SCRIPT “DESIGNOTA_TEST.M”, REPORT YOUR SCRIPT. REPORT THE OUTPUT OF THE COMMAND WINDOW, COMPARE THE RESULTS TO THE RESULTS OBTAINED BY ADT DEVICE XPLORE.

```
% OTA Design Script
% Write the SPECS
clear all;
AVDC = 34 ;
GBW = 100e6 ;
CL = 500e-15 ;
specs = struct('AVDC', AVDC,...
'CL', CL,...
'GBW', GBW);
OTA = designOTA(specs);
% Print the solution
fprintf('**** OTA Design ****\n\n');
fprintf('Input Pair:\n');
fprintf('      L = %.2f um\n      W=%.2f um\n      ViCM=%.4f\n\n',OTA.M1.L,OTA.M1.W,OTA.M1.VG);
fprintf('CM Load:\n');
fprintf('      L = %.2f um\n      W=%.2f\n\n',OTA.M3.L,OTA.M3.W);
fprintf('Tail Current Source:\n');
fprintf('      L = %.2f um\n      W=%.2f\n\n',OTA.M5.L,OTA.M5.W);
```

Input Pair:

L = 0.28 um
W=3.01 um
ViCM=1.1928 V

CM Load:

L = 1.70 um
W=31.44 um

Tail Current Source:

L = 1.00 um
W=9.06 um

		ADT	SCRIPT
M1(INPUT)	L (um)	0.28	0.28
	W (um)	3.01	3.01
M3(Load)	L (um)	1.7	1.7
	W (um)	31.38	31.44
M5(TAIL)	L (um)	1	1
	W (um)	9	9.06

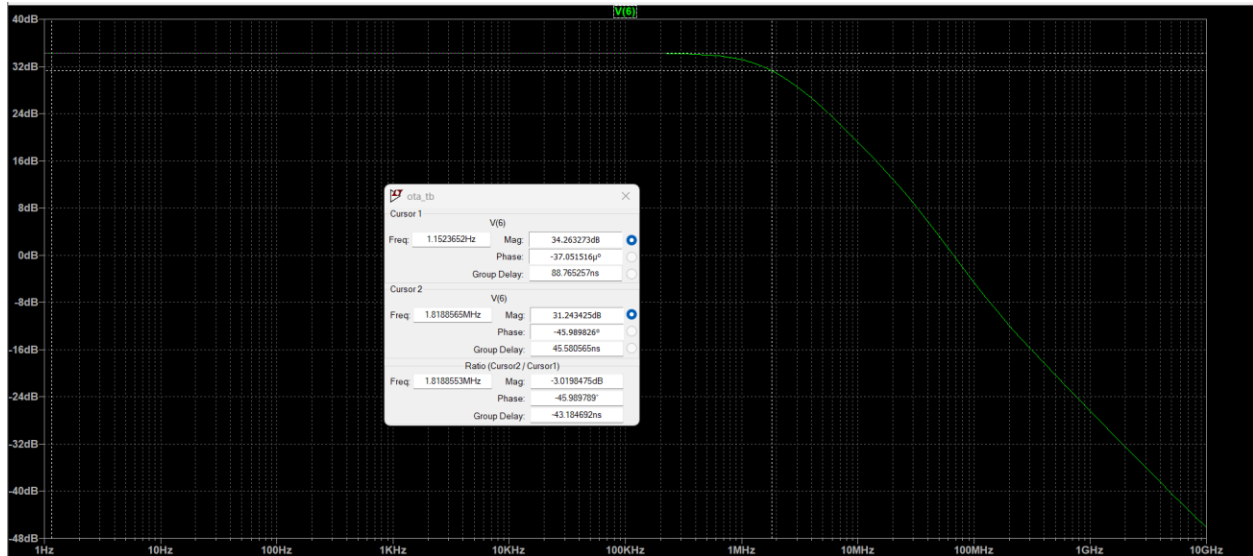
The results are almost the same, just second-degree differences because of small approximations.

7 COMPLETE THE NETLIST “OTA_TB.CIR”, SIMULATE THE DIFFERENTIAL GAIN VS FREQUENCY (ANNOTATE DC GAIN AND GBW), REPORT YOUR NETLIST, THE SIMULATION RESULTS, AND THE COMPARISON TABLE. COMMENT ON THE RESULTS.

```
5T OTA
* Include external file that contains MOSFET Model
.INCLUDE ee214b_hspice.sp
** Circuit Description **
* power supply
VDD 7 0 DC 1.8
* input
VAC1 3 8 ac 0.5
VAC2 4 9 ac -0.5
VDC1 8 0 dc 1.1928
VDC2 9 0 dc 1.1928

* circuit
* 5T OTA
M1 6 4 2 0 nch L=0.28u W=3.01u
M2 5 3 2 0 nch L=0.28u W=3.01u
M3 6 5 7 7 pch L=1.7u W=31.38u
M4 5 5 7 7 pch L=1.7u W=31.38u
M5 2 1 0 0 nch L=1u W=18u
CL 6 0 500f
* Current Mirror
M6 1 1 0 0 nch L=1u W=9u
Iref 7 1 21u
** Analysis Requests **
.op
.ac dec 10 1 10e9
** Outputs Requests **
*.PROBE

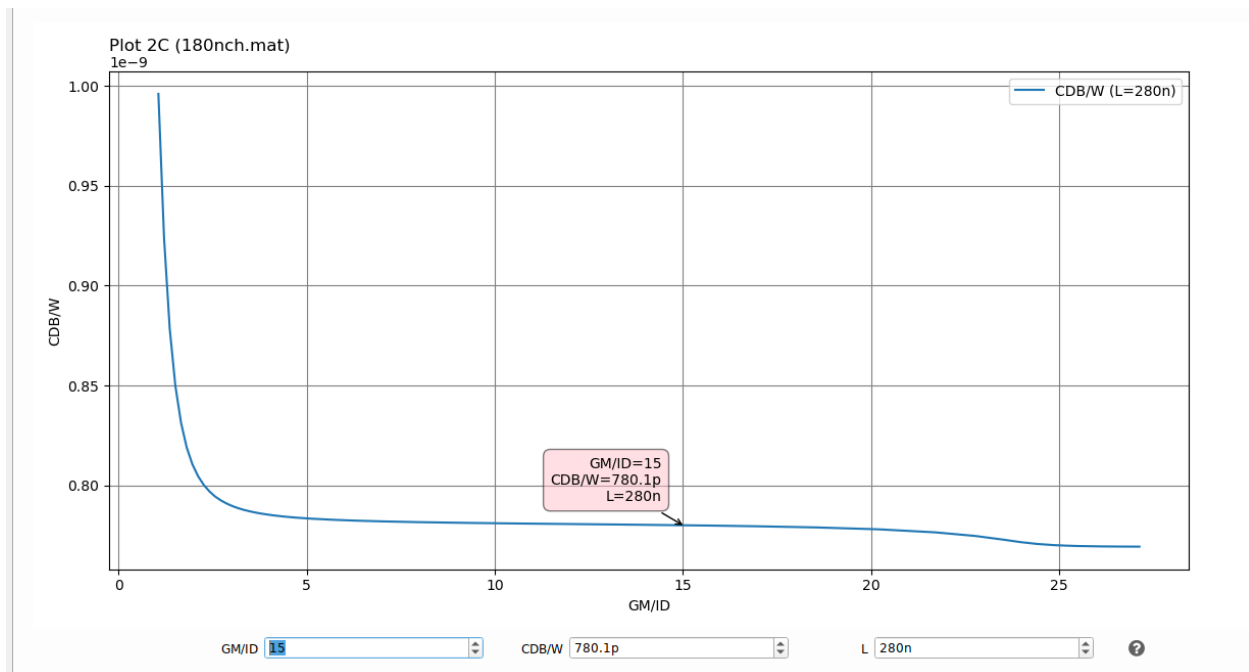
.END
```



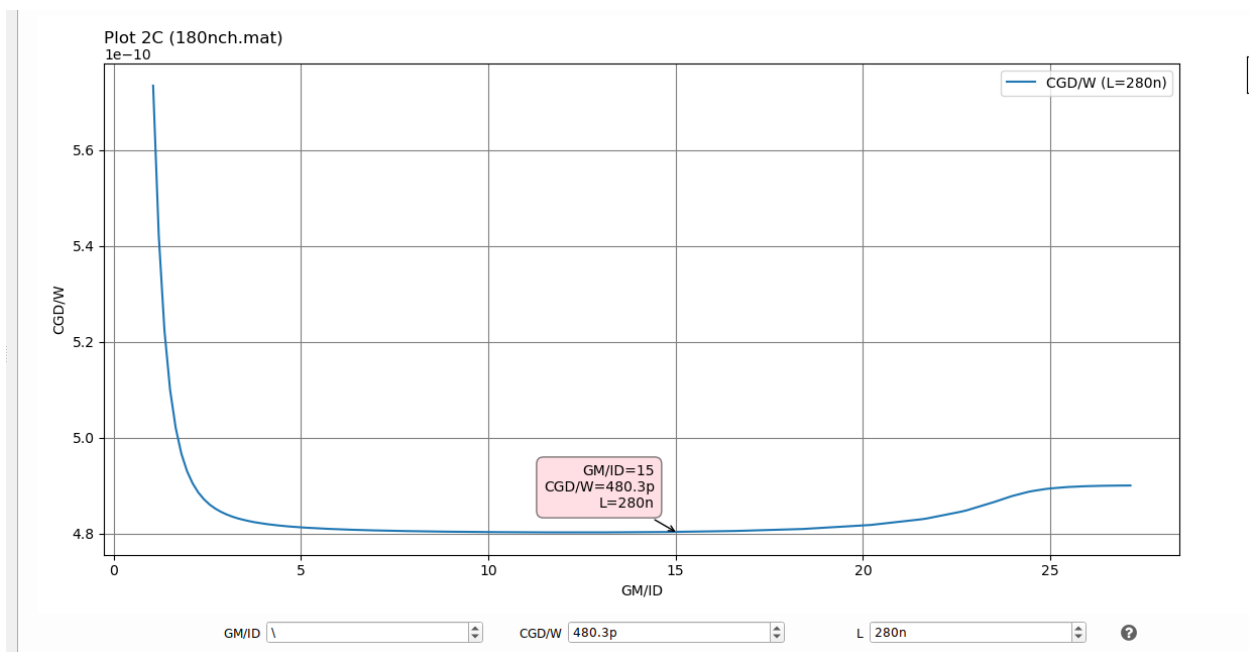
	SPICE	SPECS
GBW	$1.818 \times 51.64 = 93.88$ MHZ	100 MHZ
GAIN	34.26 DB	34 DB

COMMENT:

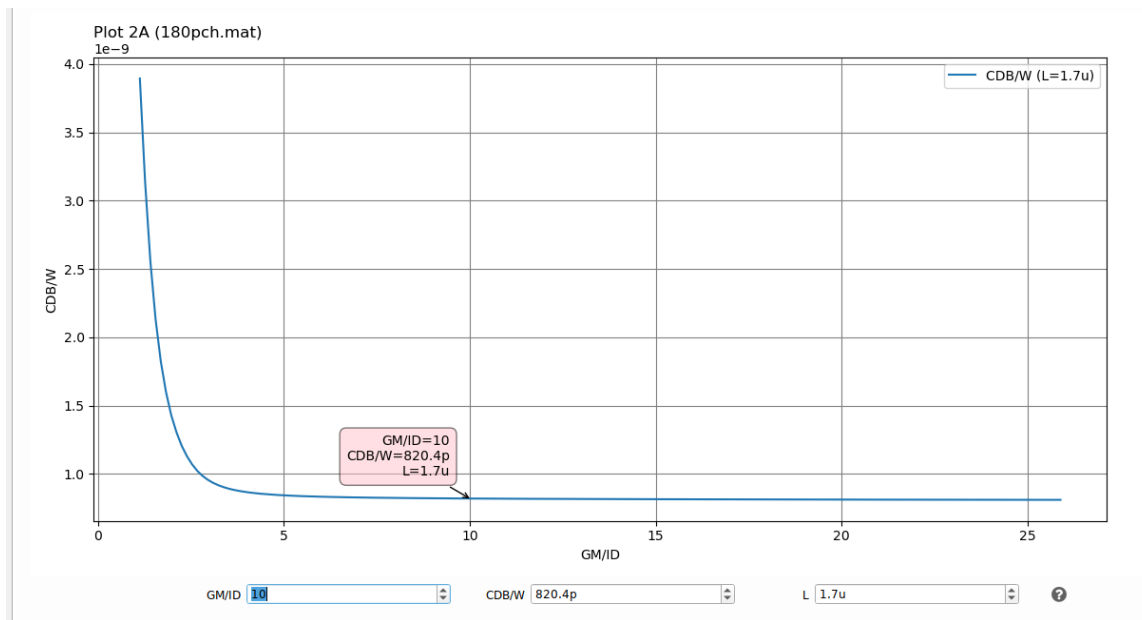
GBW is less than 100 MHZ because of the self-parasitic capacitances that we didn't consider.



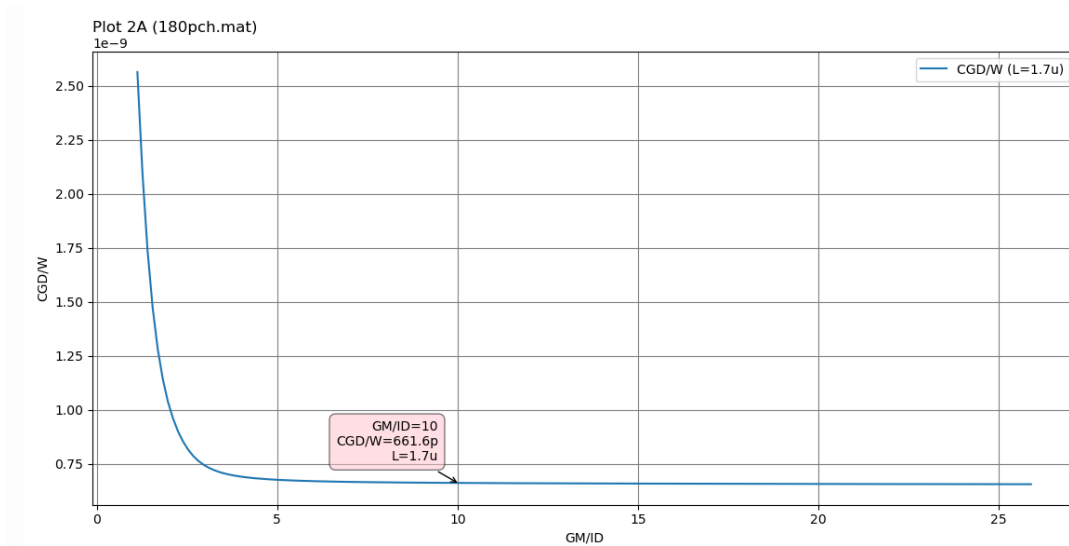
$$C_{DBin} = 2.3fF$$



$$C_{DBin} = 1.45fF$$



$$C_{DBload} = 25.74fF$$



$$C_{GDload} = 20.8fF$$

$$CL_{EFF} = CL + C_{DBin} + C_{GDin} + C_{DBload} + C_{GDload} = 550.29fF$$

$$GBW = \frac{gm}{2\pi CL_{eff}} = 91MHz, \text{ I didn't consider body effect.}$$

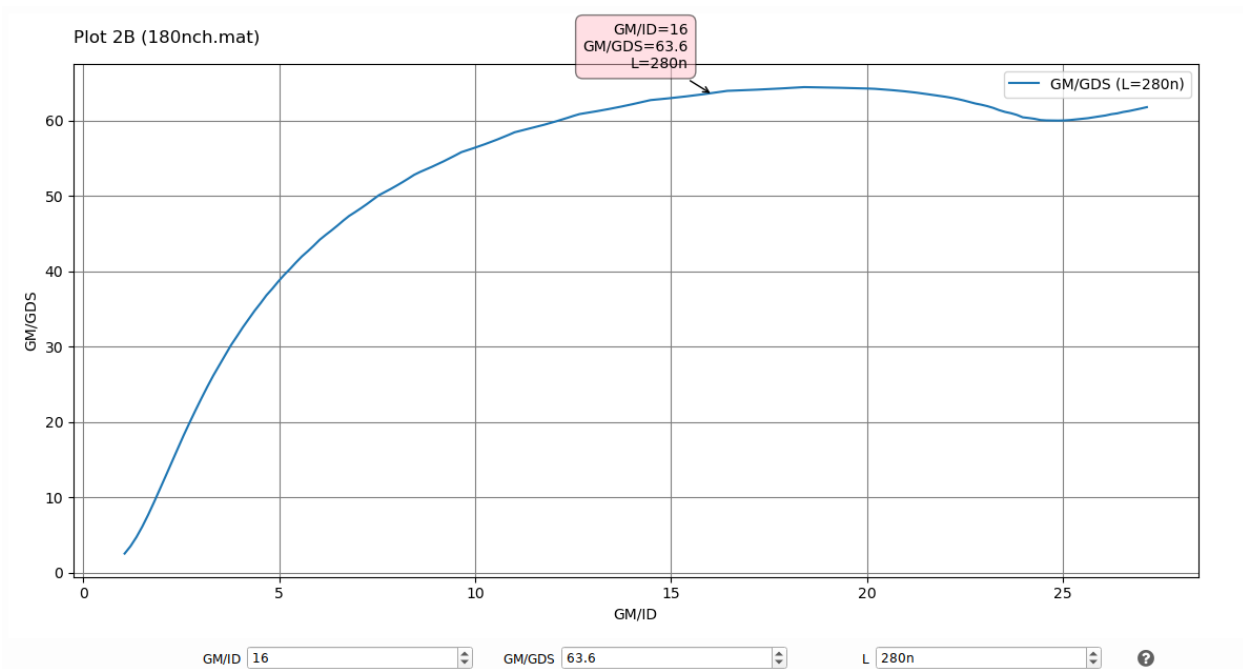
PART2

1 USE ADT DEVICE XPLORE TO REDESIGN THE PREVIOUS OTA WHILE TAKING INTO ACCOUNT THE OTA SELF-LOADING , COMPARE THE RESULTS TO THE RESULTS OBTAINED IN PART 1

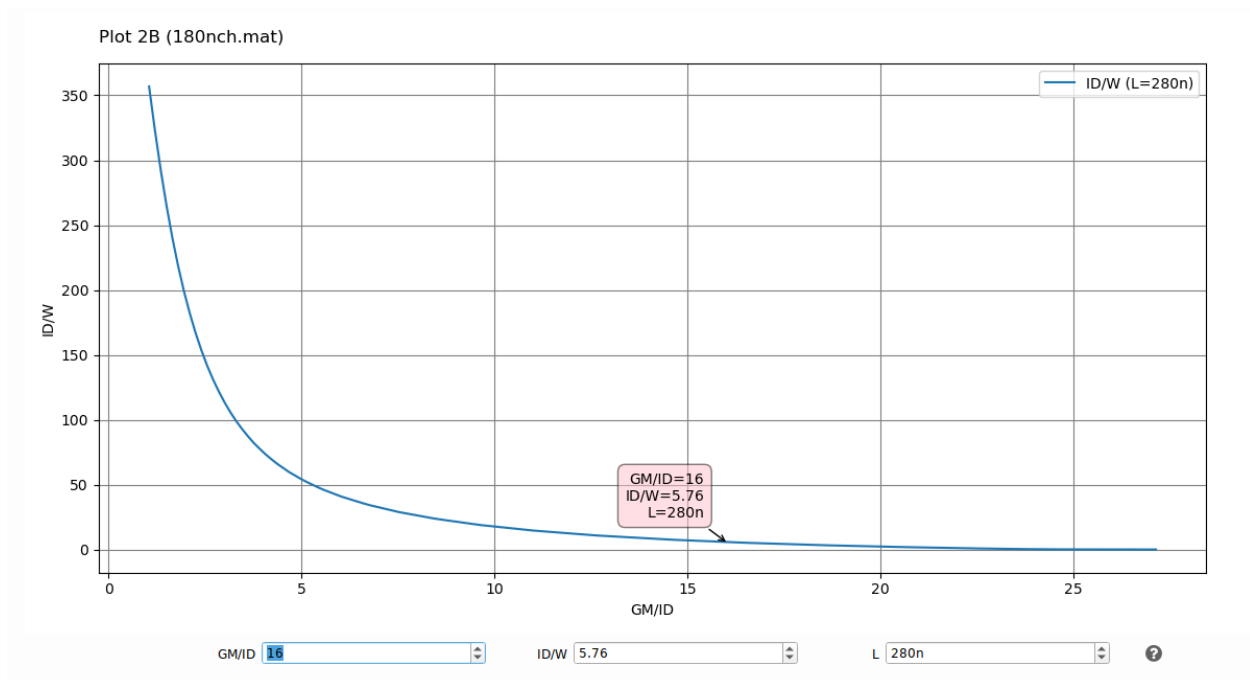
To get less current, so PMOS transistor would have less size, we need to bias the input transistor more into WI, so we choose:

$$\frac{gm}{id} = 16, \text{ as } CL_{eff} = 550.29fF, GBW = \frac{gm_{in}}{2\pi CL_{eff}}, gm_{in} = 345.75 \mu S$$

we choose $gm_{in}=350 \mu S$, so $I_d=21.875 \mu A$



$$\frac{gm}{gds} = 63.6, gds_{in} = 5.5 \mu S$$

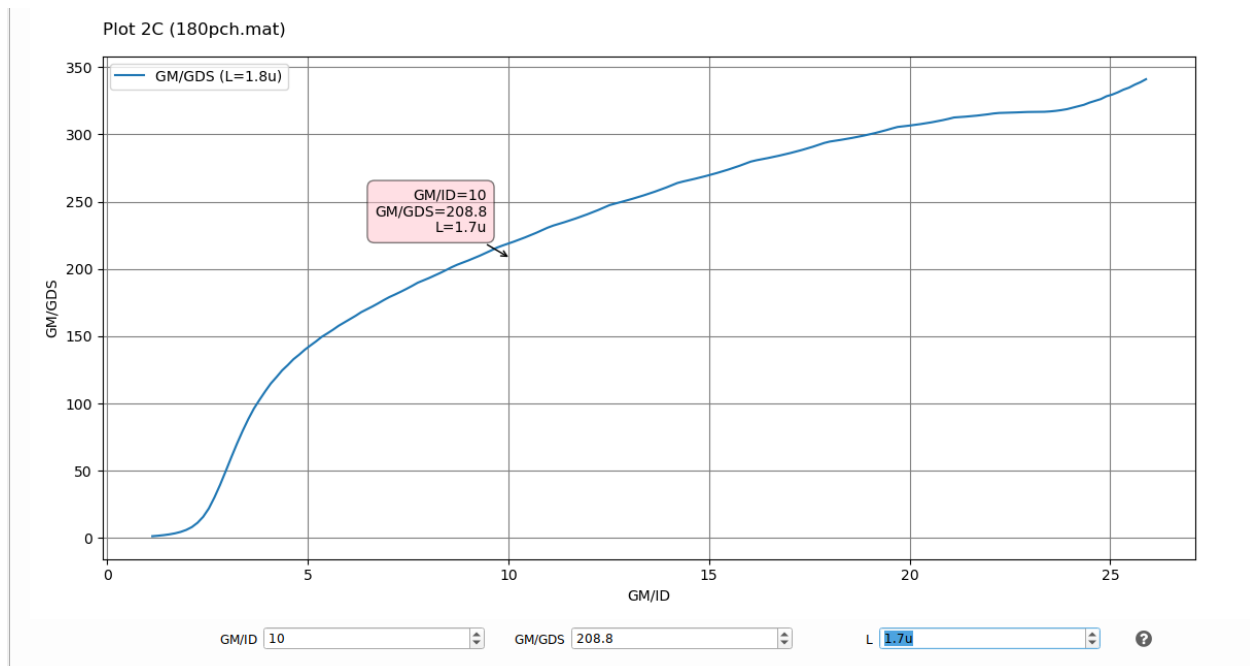


W=3.8u

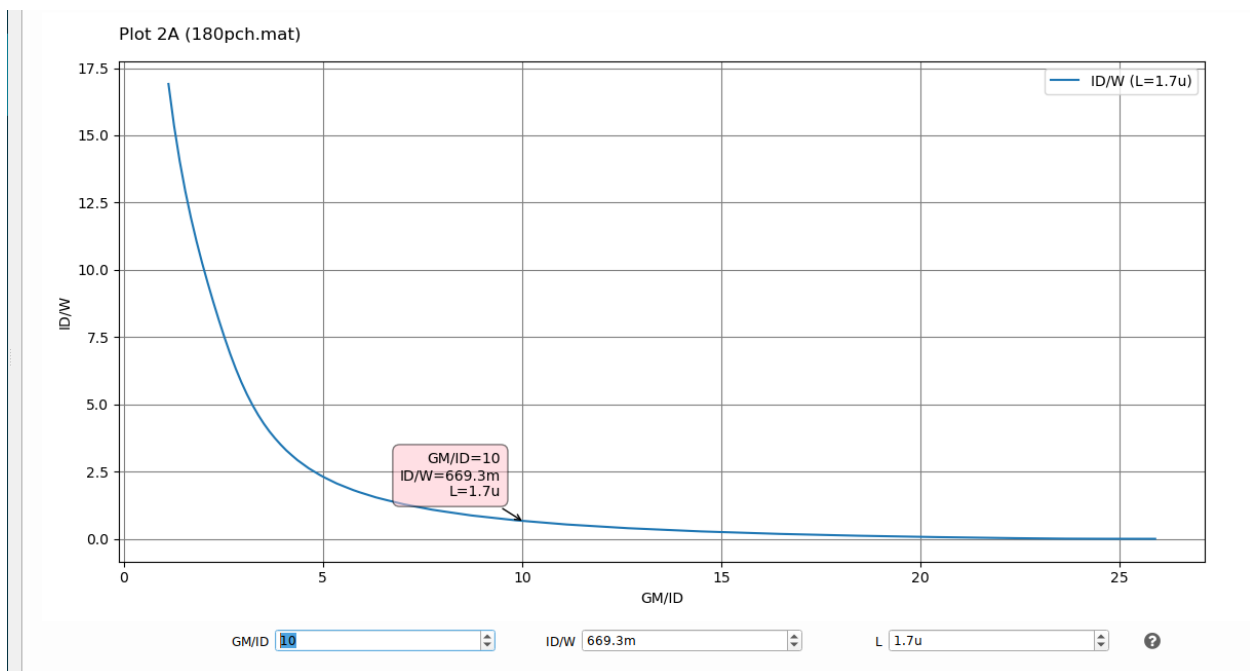
We have slightly bigger W but anyway nmos caps weren't the main factor of GBW decreasing.

$$\left(\frac{gm}{id}\right)_{load} = 10, gm = 218.75\mu S$$

$$gds_{load} = 1.1\mu S, \left(\frac{gm}{gds}\right)_{load} = 198.86$$

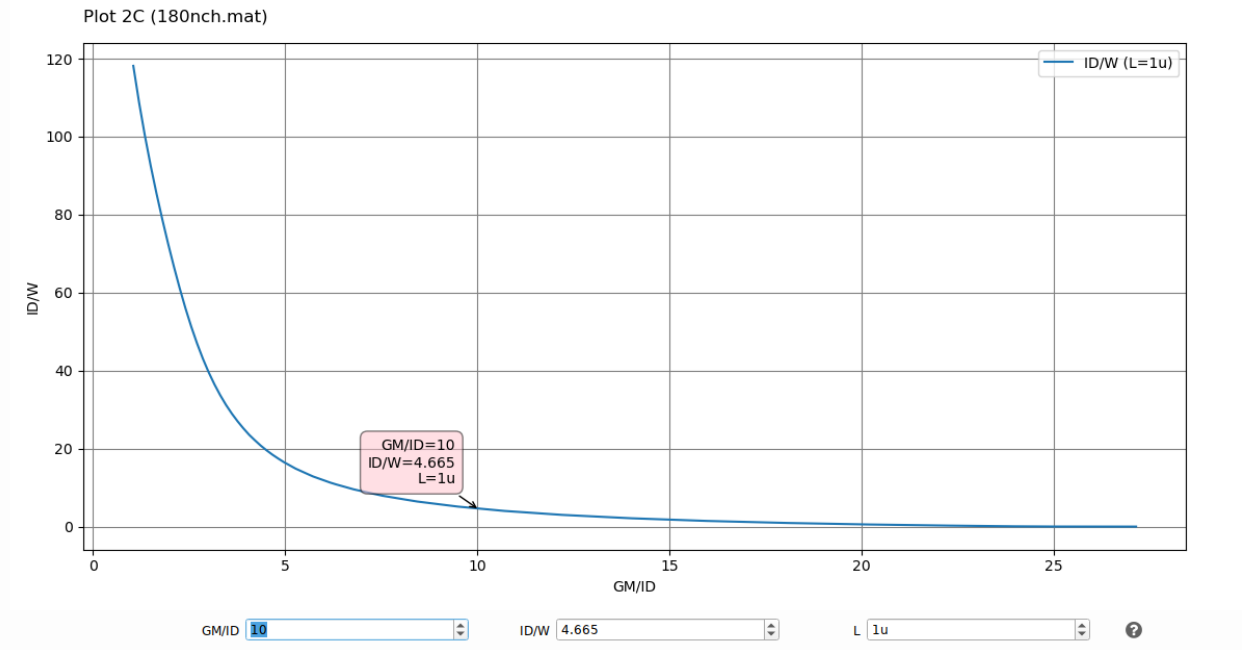


So we choose **L=1.7u** to get less W



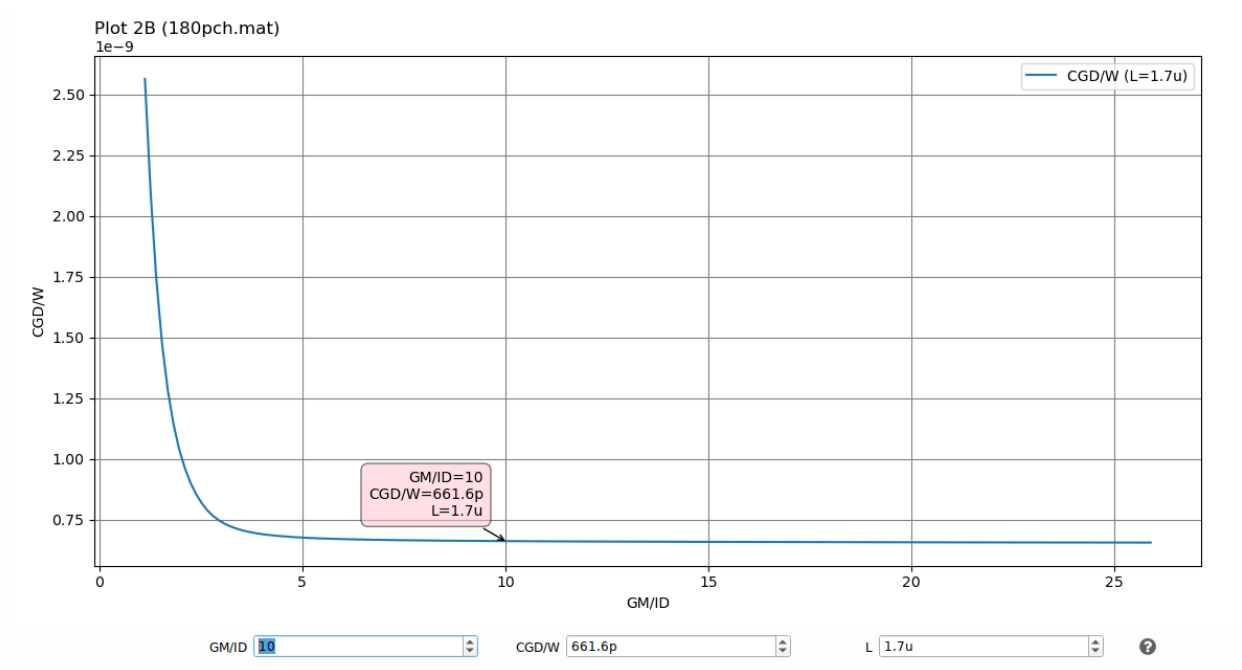
$$\frac{ID}{W} = 669.3m, W_{load} = 32.7 \mu m$$

For tail we have $I=43.75$ mA

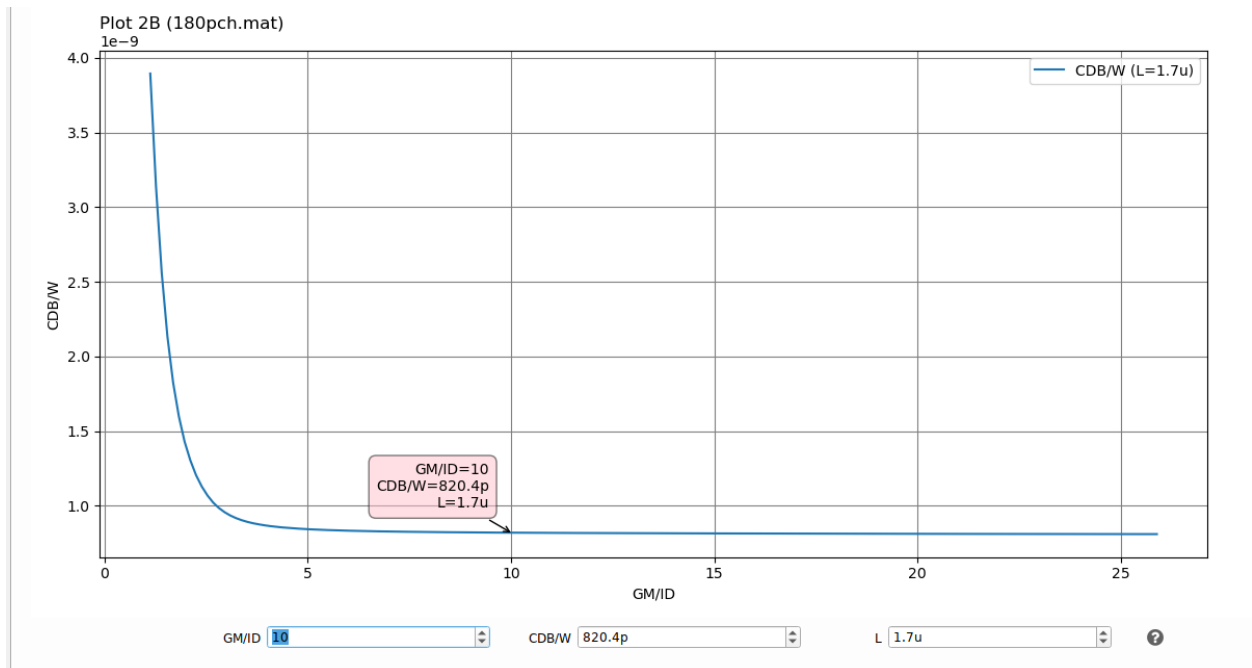


$W=9.37$ μ m

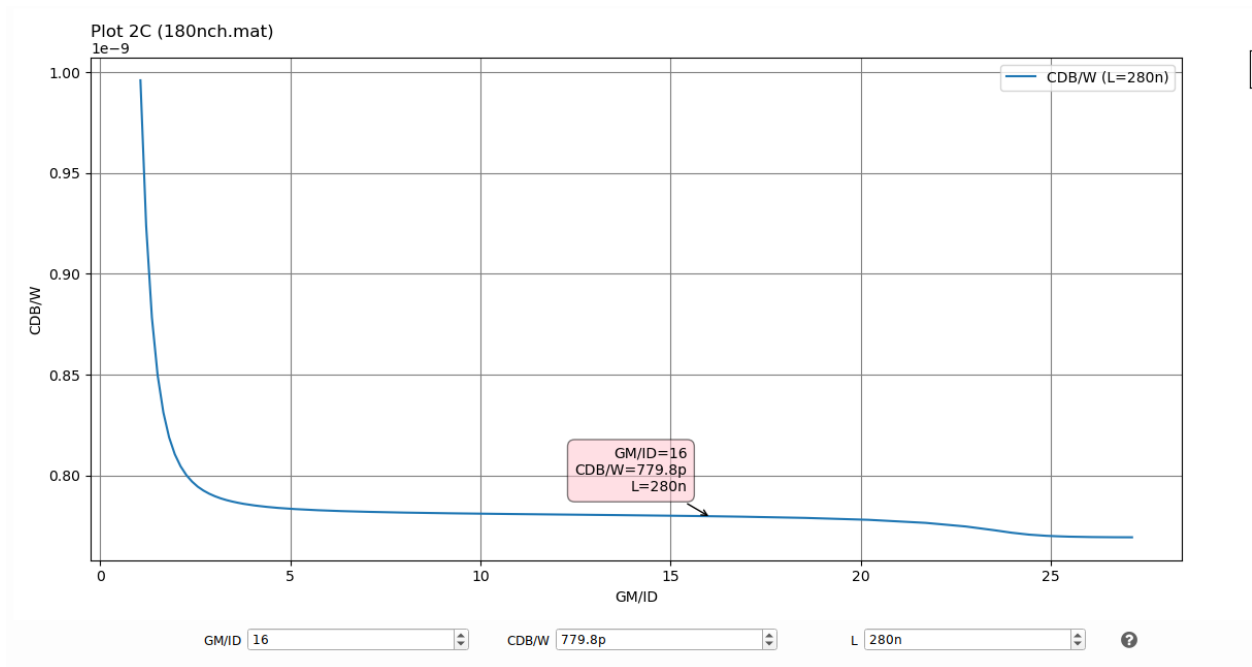
Calculate load capacitance again:



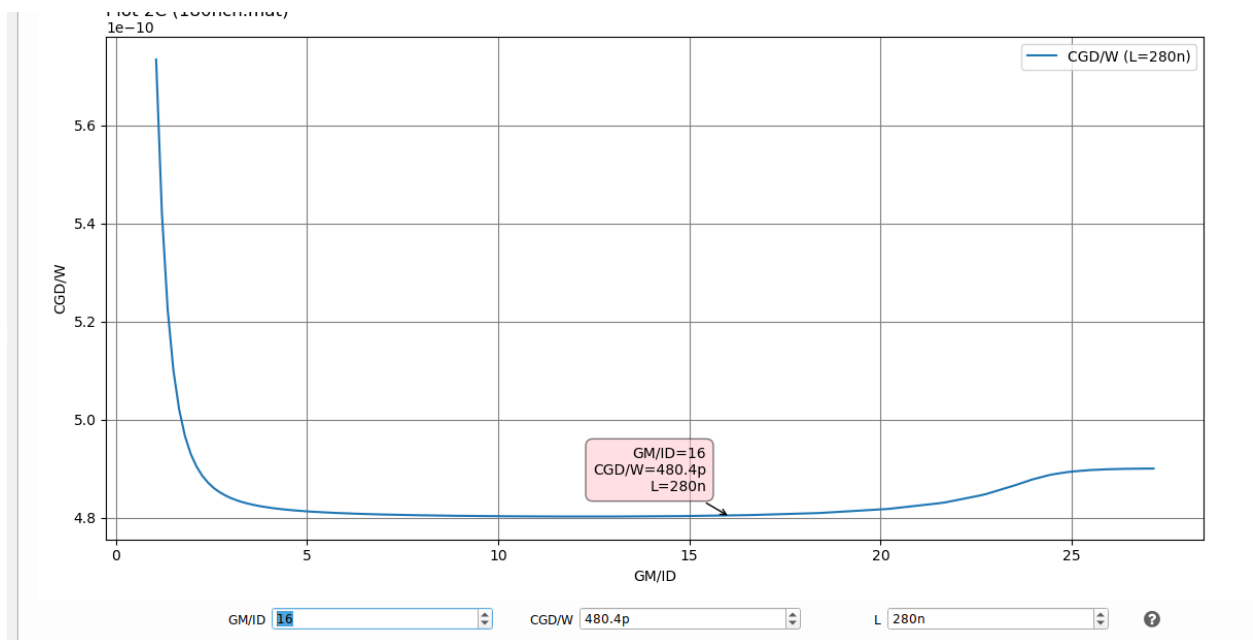
$CGD_{LOAD} = 21.63$ fF



$$CDB_{LOAD} = 26.8fF$$



$$CDB_{IN} = 2.96fF$$



$$CGD_{IN} = 1.82fF$$

$$CL_{eff} = 553.21fF, gm = 350 S$$

$$GBW = \frac{gm}{2\pi CL_{eff}} = 100.7MHz$$

1.1 COMPARISION WITH PART1:

		PART1	PART2
M1(INPUT)	L (um)	0.28	0.28
	W (um)	3.01	3.8
M3(LOAD)	L (um)	1.7	1.7
	W (um)	31.38	32.7
M5(TAIL)	L (um)	1	1
	W (um)	9	9.37

2 MATLAB SCRIPT,. COMPARE THE RESULTS TO THE RESULTS OBTAINED BY ADT DEVICE XPLORE

2.1 REPORT THE OUTPUT OF THE COMMAND WINDOW

```
Input Pair:
  L = 0.28 um
  W=3.80 um
  ViCM=1.1801 V

CM Load:
  L = 1.50 um
  W=28.49 um

Tail Current Source:
  L = 1.00 um
  W=9.36 um
```

2.2 COMPARE THE RESULTS TO THE RESULTS OBTAINED IN PART 1.

		PART1	PART2
M1(INPUT)	L (um)	0.28	0.28
	W (um)	3.01	3.8
M3(Load)	L (um)	1.7	1.5
	W (um)	31.38	28.49
M5(TAIL)	L (um)	1	1
	W (um)	9	9.36

2.3 COMPARE THE RESULTS TO THE RESULTS OBTAINED BY ADT DEVICE XPLORE.

		Script	ADT
M1(INPUT)	L (um)	0.28	0.28
	W (um)	3.8	3.8
M3(LOAD)	L (um)	1.5	1.7
	W (um)	28.49	32.7
M5(TAIL)	L (um)	1	1
	W (um)	9.36	9.37

MATLAB code is more optimized due it didn't take any margin or approximation that I took.

3 WRITE A NETLIST FOR THE SYNTHESIZED OTA. SIMULATE THE DIFFERENTIAL GAIN VS FREQUENCY (ANNOTATE DC GAIN AND GBW) . COMMENT ON THE RESULTS.

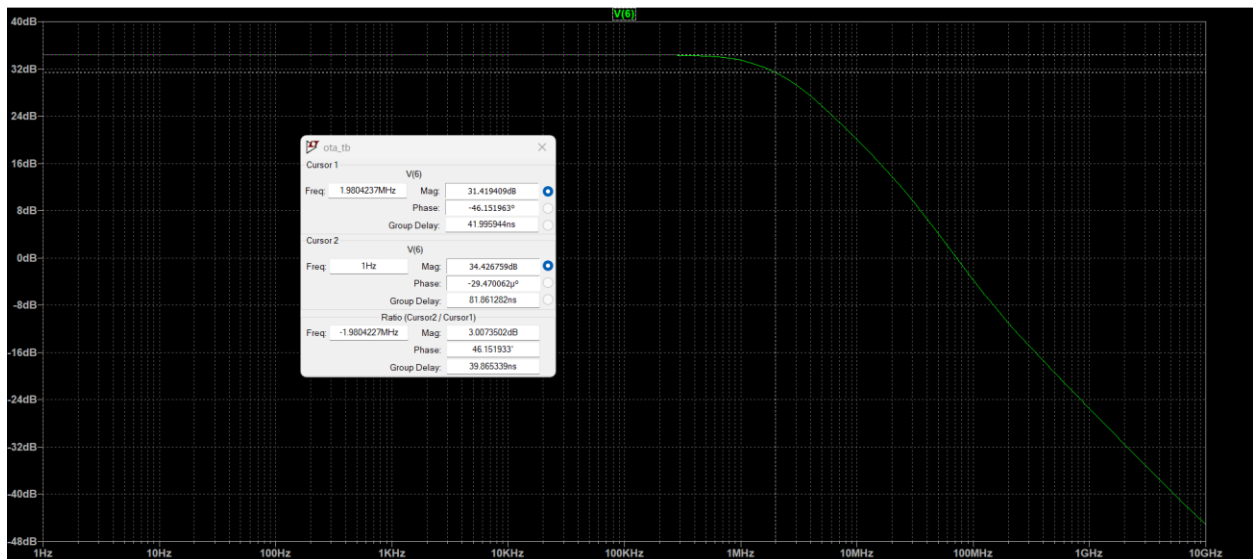
3.1 REPORT YOUR NETLISTS

```
5T OTA
* Include external file that contains MOSFET Model
.INCLUDE ee214b_hspice.sp
** Circuit Description **
* power supply
VDD 7 0 DC 1.8
* input
VAC1 3 8 ac 0.5
VAC2 4 9 ac -0.5
VDC1 8 0 dc 1.1928
VDC2 9 0 dc 1.1928

* circuit
* 5T OTA
M1 6 4 2 0 nch L=0.28u W=3.8u
M2 5 3 2 0 nch L=0.28u W=3.8u
M3 6 5 7 7 pch L=1.7u W=32.7u
M4 5 5 7 7 pch L=1.7u W=32.7u
M5 2 1 0 0 nch L=1u W=18.74u
CL 6 0 500f
* Current Mirror
M6 1 1 0 0 nch L=1u W=9.37u
Iref 7 1 21.875u
** Analysis Requests **
.op
.ac dec 10 1 10e9
** Outputs Requests **
*.PROBE

.END
```

3.2 THE SIMULATION RESULTS



GAIN=34.42 DB

GBW=52.63*1.98MHZ=104.2MHZ

3.3 COMPARISON TABLE

	SPICE	SPECS
GBW	104.2 MHZ	100 MHZ
GAIN	34.42 DB	34 DB

Results meet the specs, using lookup tables was very helpful and practical for design.